Eleanor Pike

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Education

PhD in Astrophysics (Expected thesis submission: March 2027) 2023 – Present

University of Sheffield, UK

Supervisor: Prof. S. Littlefair

Expected thesis title: *Spectroscopic Variability and Auroral Chemistry in Ultracool Dwarf Atmospheres*

MSci (Hons) in Physics and Astronomy (First-Class Honours) 2018 – 2023

University of Glasgow, UK

Supervisor: Prof. L. Fletcher

Final year project title: *Hot Flows in Solar Active Regions from Hinode EIS*

Research Experience

PhD Research

My PhD combines observational and computational approaches to investigate ultracool dwarf atmospheres, exploring the potential relationships between auroral activity, atmospheric chemistry and spectroscopic variability.

- Adapted and extended early JWST/NIRISS-SOSS data reduction pipelines, modifying spectral tracing and extraction procedures to address limitations in the available software.
- Investigated spectroscopic variability using principal component analysis and comparisons with optical Hα observations.
- Performed atmospheric forward modelling and used radiative transfer software (PICASO) to investigate the effects of temperature structure and chemical composition on emergent spectra.
- Developed an exploratory MCMC-based atmospheric retrieval framework for parameter estimation from spectroscopic observations.
- Developing and adapting non-equilibrium chemical kinetics models (STAND/ARGO), including treatments of condensation, to investigate the effects of auroral electron precipitation on atmospheric composition.

Member of the ULTRACAM Team

- Participated in observing runs with ULTRACAM, ULTRASPEC and HiPERCAM, gaining experience in the operation of high-speed astronomical photometry instruments.
- Prepared nightly observing schedules based on allocated telescope time and scientific observing priorities.
- Supported instrument operations during observing runs and produced nightly and post-run reports documenting observations and observing conditions.

Teaching Experience

Graduate Teaching Assistant 2024 – Present

- Supported first- and second-year undergraduate students in computational and observational astronomy laboratories.
- Guided students in Python programming, computational problem-solving and the reduction and analysis of astronomical observations.
- Provided guidance on scientific writing and assessed laboratory reports, programming assignments and presentations, giving constructive feedback.

Organiser, Astronomy Early Career Researcher Sessions 2024 – 2025

- Organised and facilitated weekly meetings for astronomy postgraduate students and postdoctoral researchers, identifying shared interests and areas for skills development.
- Planned and delivered sessions and coordinated contributions from other researchers.

Skills

- **Scientific programming:** Proficient in Python; experience with Fortran, SQL/ADQL, debugging, numerical modelling and model validation.
- **Computing tools:** Linux and macOS; Git/GitHub for version control and scientific software development.
- **Science communication:** Best Physics and Astronomy Poster at the University of Sheffield Faculty of Science poster showcase; participated in sixth-form outreach encouraging women into STEM.

Publications

First-author papers

- E. Pike et al. (2026). **"JWST/NIRISS-SOSS Spectroscopic Variability and Auroral Modelling of Warm Ultracool Dwarf LSR J1835+3259."** *MNRAS* (under review).
- E. Pike et al. **"The Chemical Impact of Auroral Electron Beams on Ultracool Dwarf Atmospheres."** (in preparation).

Selected co-author papers [1]

- F. Wang et al. (2026). **"Auroral Heating and Chemistry in the Warm Ultracool Dwarf LSR J1835+3259 I: Extreme-State Atmospheric Retrievals with JWST/NIRISS-SOSS."** *MNRAS* (under review).
- F. Wang et al. (2026). **"Auroral Heating and Chemistry in the Warm Ultracool Dwarf LSR J1835+3259 II: Time-Resolved Retrievals from JWST/NIRISS-SOSS."** *MNRAS* (under review).
- A. Yates et al. (2026). **"Long-term eclipse time variations in white dwarf binaries."** *MNRAS*, 547(2), 1-14.
- N. Castro Segura et al. (2025). **"A sibling of AR Scorpii: SDSS J230641.47+244055.8 and the observational blueprint of white dwarf pulsars."** *MNRAS*, 543, 2116–2129.
- H. S. Hudson et al. (2022). **"Fast prograde coronal flows in solar active regions."** *MNRAS Letters*, 515, L84–L88.

Conference Contributions

- **"JWST Spectroscopy and Auroral Modelling of Warm UCD LSR J1835+3259."**
  Contributed talk. Substellar Astrophysics, Tordesillas, 10–14 August 2026.
- **"JWST Spectroscopic Variability and Auroral Modelling of Warm Ultracool Dwarf LSR J1835+3259."**
  Contributed talk. National Astronomy Meeting, Birmingham, 20–24 July 2026.
- **"JWST/NIRISS Observations of an Auroral Brown Dwarf."**
  Contributed talk. Cool Stars 23, Tokyo, 15–19 June 2026.
- **"JWST Observations of an Auroral Brown Dwarf."**
  Contributed talk. European Astronomical Society Meeting, Cork, 23–27 June 2025.
- **"JWST Observations of Aurora on a Warm Brown Dwarf."**
  Contributed talk. Cool Stars 22, San Diego, 24–28 June 2024.

[1] Full publication list is available on my website